

Clustering Assignment Solutions

Question 1: K-means Algorithm (6 points)

Given data points

- $s_1 = (6, 11)$
- $s_2 = (3, 4)$
- $s_3 = (8, 6)$
- $s_4 = (9, 7)$
- $s_5 = (7, 10)$
- $s_6 = (10, 6)$
- $s_7 = (4, 7)$
- $s_8 = (4, 12)$

Initial centers

- Cluster 1 center: $s_2(3, 4)$
- Cluster 2 center: $s_5(7, 10)$
- Cluster 3 center: $s_8(4, 12)$

First Round of Execution

Step 1: Assign to nearest center

- $s_1 = (6, 11)$:

$$d(s_1, C_1) = \sqrt{(6 - 3)^2 + (11 - 4)^2} = \sqrt{9 + 49} = \sqrt{58} \approx 7.62$$

$$d(s_1, C_2) = \sqrt{(6 - 7)^2 + (11 - 10)^2} = \sqrt{1 + 1} = \sqrt{2} \approx 1.41$$

$$d(s_1, C_3) = \sqrt{(6 - 4)^2 + (11 - 12)^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.24$$

$\Rightarrow$ **assign** s_1 to Cluster 2.

- $s_2 = (3, 4)$:

$$d(s_2, C_1) = 0$$

$\Rightarrow$ **assign** s_2 to Cluster 1.

- $s_3 = (8, 6)$:

$$d(s_3, C_1) = \sqrt{(8 - 3)^2 + (6 - 4)^2} \approx 5.39$$

$$d(s_3, C_2) = \sqrt{(8 - 7)^2 + (6 - 10)^2} \approx 4.12$$

$$d(s_3, C_3) = \sqrt{(8 - 4)^2 + (6 - 12)^2} \approx 7.21$$

$\Rightarrow$ **assign** s_3 to Cluster 2.

- $s_4 = (9, 7)$:

$$d(s_4, C_1) = \sqrt{(9 - 3)^2 + (7 - 4)^2} \approx 6.71$$

$$d(s_4, C_2) = \sqrt{(9 - 7)^2 + (7 - 10)^2} \approx 3.61$$

$$d(s_4, C_3) = \sqrt{(9 - 4)^2 + (7 - 12)^2} \approx 7.07$$

$\Rightarrow$ **assign** s_4 to Cluster 2.

- $s_5 = (7, 10)$:

$$d(s_5, C_2) = \sqrt{(7 - 7)^2 + (10 - 10)^2} = 0$$

$\Rightarrow$ **assign** s_5 to Cluster 2.

- $s_6 = (10, 6)$:

$$d(s_6, C_1) = \sqrt{(10 - 3)^2 + (6 - 4)^2} \approx 7.28$$

$$d(s_6, C_2) = \sqrt{(10 - 7)^2 + (6 - 10)^2} \approx 5$$

$$d(s_6, C_3) = \sqrt{(10 - 4)^2 + (6 - 12)^2} \approx 8.49$$

$\Rightarrow$ **assign** s_6 to Cluster 2.

- $s_7 = (4, 7)$:

$$d(s_7, C_1) = \sqrt{(4 - 3)^2 + (7 - 4)^2} \approx 3.16$$

$$d(s_7, C_2) = \sqrt{(4 - 7)^2 + (7 - 10)^2} \approx 4.24$$

$$d(s_7, C_3) = \sqrt{(4 - 4)^2 + (7 - 12)^2} \approx 5$$

$\Rightarrow$ **assign** s_7 to Cluster 1.

- $s_8 = (4, 12)$:

$$d(s_8, C_3) = \sqrt{(4 - 4)^2 + (7 - 12)^2} \approx 0$$

$\Rightarrow$ **assign** s_8 to Cluster 3.

Step 2: Find new centers

After First Round:

- **Cluster 1:** $s_2(3, 4), s_7(4, 7)$
 - New center: $((3 + 4)/2, (4 + 7)/2) = (3.5, 5.5)$
- **Cluster 2:** $s_1(6, 11), s_3(8, 6), s_4(9, 7), s_5(7, 10), s_6(10, 6)$
 - New center: $((6 + 8 + 9 + 7 + 10)/5, (11 + 6 + 7 + 10 + 6)/5) = (8, 8)$
- **Cluster 3:** $s_8(4, 12)$
 - New center: $(4, 12)$

Second Iteration

Using new centers: $C_1(3.5, 5.5), C_2(8, 8), C_3(4, 12)$

Reassignment:

- **Cluster 1:** $s_2(3, 4), s_7(4, 7)$
- **Cluster 2:** $s_3(8, 6), s_4(9, 7), s_6(10, 6)$
- **Cluster 3:** $s_1(6, 11), s_5(7, 10), s_8(4, 12)$

New centers:

- **Cluster 1:** $(3.5, 5.5)$
- **Cluster 2:** $((8 + 9 + 10)/3, (6 + 7 + 6)/3) = (9, 6.33)$
- **Cluster 3:** $((6 + 7 + 4)/3, (11 + 10 + 12)/3) = (5.67, 11)$

Third Iteration (Final)

Using centers: $C_1(3.5, 5.5), C_2(9, 6.33), C_3(5.67, 11)$

Final assignment:

- **Cluster 1:** $s_2(3, 4), s_7(4, 7)$
 - Final center: $(3.5, 5.5)$
- **Cluster 2:** $s_3(8, 6), s_4(9, 7), s_6(10, 6)$
 - Final center: $(9, 6.33)$
- **Cluster 3:** $s_1(6, 11), s_5(7, 10), s_8(4, 12)$
 - Final center: $(5.67, 11)$

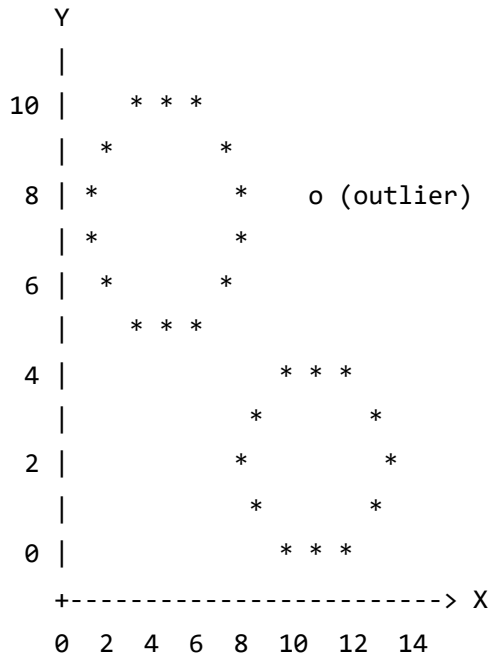
Question 2: Clustering Method Suitability (4 points)

Density-based Clustering (DBSCAN) vs. Partitioning-based (K-means)

Conditions under which DBSCAN is more suitable over K-means:

1. **Non-spherical clusters:** DBSCAN can find clusters of any shapes
2. **Cluster densities:** DBSCAN handles clusters with different densities
3. **Outliers:** DBSCAN looks for and excludes outliers
4. **Unknown number of clusters:** DBSCAN doesn't need to pre-specify k

Scenario favoring DBSCAN



In this scenario, there are two oval shaped clusters with an outlier.

DBSCAN, in this scenario, would correctly identify both ovals.

Meanwhile, K-means would split these two into spherical regions.

K-means assumes spherical clusters had equal densities.

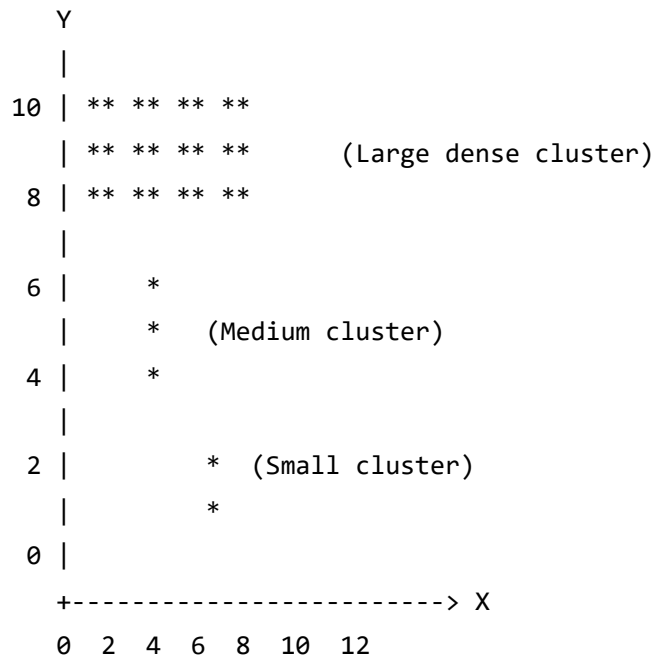
Hierarchical Clustering vs. Partitioning-based (K-means)

Conditions where Hierarchical Clustering is more suitable over K-means:

1. **Cluster hierarchy:** When you need to see how clusters form smaller subclusters
2. **No predetermined k:** When the number of clusters is unknown
3. **Dendrogram visualization:** When you want a full picture of how clusters merge and relate
4. **Small to medium datasets:** Hierarchical clustering is computationally expensive for large datasets
5. **Varying cluster sizes:** When clusters have different sizes

Graphical Illustration:

Scenario favoring Hierarchical Clustering



Hierarchical clustering naturally identifies the large, medium, and small clusters without forcing them into similar sizes. A dendrogram also shows how the clusters merge step by step. On the other hand, K-means is bad with clusters of different sizes and could split the large cluster or force all groups to be more similar when they're not.